11 Random Graphs

At various points in this book we already encountered the following fundamental theorem of Erdős: *for every integer k there is a graph G with $g(G) > k$ and $\chi(G) > k$.* In plain English: there exist graphs combining arbitrarily large girth with arbitrarily high chromatic number.

How could one prove such a theorem? The standard approach would be to construct a graph with those two properties, possibly in steps by induction on k. However, this is anything but straightforward: the global nature of the second property forced by the first, namely, that the graph should have high chromatic number 'overall' but be acyclic (and hence 2-colourable) locally, flies in the face of any attempt to build it up, constructively, from smaller pieces that have the same or similar properties.

In his pioneering paper of 1959, Erdős took a radically different approach: for each n he defined a probability space on the set of graphs with n vertices, and showed that, for some carefully chosen probability measures, the probability that an n-vertex graph has both of the above properties is positive for all large enough n.

This approach, now called the *probabilistic method*, has since unfolded into a sophisticated and versatile proof technique, in graph theory as much as in other branches of discrete mathematics. The theory of *random graphs* is now a subject in its own right. The aim of this chapter is to offer an elementary but rigorous introduction to random graphs: no more than is necessary to understand its basic concepts, ideas and techniques, but enough to give an inkling of the power and elegance hidden behind the calculations.

Erdős's theorem asserts the existence of a graph with certain properties: it is a perfectly ordinary assertion showing no trace of the randomness employed in its proof. There are also results in random graphs that are generically random even in their statement: these are theorems about *almost all* graphs, a notion we shall meet in Section 11.3. In the

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last section, we give a detailed proof of a theorem of Erdős and Rényi that illustrates a proof technique frequently used in random graph theory, the so-called *second moment method*.

11.1 The notion of a random graph

V $\mathcal{G}$

Let V be a fixed set of n elements, say $V = \{0, \ldots, n-1\}$. Our aim is to turn the set $\mathcal{G}$ of all graphs on V into a probability space, and then to consider the kind of questions typically asked about random objects: What is the probability that a graph $G \in \mathcal{G}$ has this or that property? What is the expected value of a given invariant on G, say its expected girth or chromatic number?

p q

Intuitively, we should be able to generate G randomly as follows. For each $e \in [V]^2$ we decide by some random experiment whether or not e shall be an edge of G; these experiments are performed independently, and for each the probability of success – i.e. of accepting e as an edge for G – is equal to some fixed[1] number $p \in [0,1]$. Then if G_0 is some fixed graph on V, with m edges say, the elementary event $\{G_0\}$ has a probability of $p^m q^{\binom{n}{2}-m}$ (where $q := 1-p$): with this probability, our randomly generated graph G is this particular graph G_0. (The probability that G is *isomorphic* to G_0 will usually be greater.) But if the probabilities of all the elementary events are thus determined, then so is the entire probability measure of our desired space $\mathcal{G}$. Hence all that remains to be checked is that such a probability measure on $\mathcal{G}$, one for which all individual edges occur independently with probability p, does indeed exist.[2]

Ω_e $\mathbb{P}_e$ $\mathcal{G}(n,p)$

In order to construct such a measure on $\mathcal{G}$ formally, we start by defining for every potential edge $e \in [V]^2$ its own little probability space $\Omega_e := \{0_e, 1_e\}$, choosing $\mathbb{P}_e(\{1_e\}) := p$ and $\mathbb{P}_e(\{0_e\}) := q$ as the probabilities of its two elementary events. As our desired probability space $\mathcal{G} = \mathcal{G}(n,p)$ we then take the product space

Ω

$$\Omega := \prod_{e \in [V]^2} \Omega_e \,.$$

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[1] Often, the value of p will depend on the cardinality n of the set V on which our random graphs are generated; thus, p will be the value $p = p(n)$ of some function $n \mapsto p(n)$. Note, however, that V (and hence n) is fixed for the definition of $\mathcal{G}$: for each n separately, we are constructing a probability space of the graphs G on $V = \{0, \ldots, n-1\}$, and within each space the probability that $e \in [V]^2$ is an edge of G has the same value for all e.

[2] Any reader ready to believe this may skip ahead now to the end of Proposition 11.1.1, without missing anything.

$\mathbb{P}$

Thus, formally, an element of Ω is a map ω assigning to every $e \in [V]^2$ either 0_e or 1_e, and the probability measure $\mathbb{P}$ on Ω is the product measure of all the measures $\mathbb{P}_e$. In practice, of course, we identify ω with the graph G on V whose edge set is

$$E(G) = \{\, e \mid \omega(e) = 1_e \,\}\,,$$

random graph

and call G a *random graph* on V with edge probability p.

event

Following standard probabilistic terminology, we may now call any set of graphs on V an *event* in $\mathcal{G}(n,p)$. In particular, for every $e \in [V]^2$ the set

A_e

$$A_e := \{\, \omega \mid \omega(e) = 1_e \,\}$$

of all graphs G on V with $e \in E(G)$ is an event: the event that e is an edge of G. For these events, we can now prove formally what had been our guiding intuition all along:

Proposition 11.1.1. *The events A_e are independent and occur with probability p.*

Proof. By definition,

$$A_e = \{1_e\} \times \prod_{e' \neq e} \Omega_{e'}\,.$$

Since $\mathbb{P}$ is the product measure of all the measures $\mathbb{P}_e$, this implies

$$\mathbb{P}(A_e) = p \cdot \prod_{e' \neq e} 1 = p\,.$$

Similarly, if $\{e_1, \dots, e_k\}$ is any subset of $[V]^2$, then

$$\begin{aligned}
\mathbb{P}\left(A_{e_1} \cap \ldots \cap A_{e_k}\right) &= \mathbb{P}\Big(\{1_{e_1}\} \times \ldots \times \{1_{e_k}\} \times \prod_{e \notin \{e_1,\dots,e_k\}} \Omega_e\Big)\\
&= p^k\\
&= \mathbb{P}(A_{e_1}) \cdots \mathbb{P}(A_{e_k})\,.
\end{aligned}$$

□

As noted before, $\mathbb{P}$ is determined uniquely by the value of p and our assumption that the events A_e are independent. In order to calculate probabilities in $\mathcal{G}(n,p)$, it therefore generally suffices to work with these two assumptions: our concrete model for $\mathcal{G}(n,p)$ has served its purpose and will not be needed again.

k

ℓ

As a simple example of such a calculation, consider the event that G contains some fixed graph H on a subset of V as a subgraph; let $|H| =: k$ and $\|H\| =: \ell$. The probability of this event $H \subseteq G$ is the product of the probabilities A_e over all the edges $e \in H$, so $\mathbb{P}\,[\, H \subseteq G \,] = p^\ell$. In contrast, the probability that H is an *induced* subgraph of G is $p^\ell q^{\binom{k}{2}-\ell}$: now the edges missing from H are required to be missing from G too, and they do so independently with probability q.

The probability p_H that G has an induced subgraph *isomorphic* to H is usually more difficult to compute: since the possible instances of H on subsets of V overlap, the events that they occur in G are not independent. However, the sum (over all k-sets $U \subseteq V$) of the probabilities $\mathbb{P}[\,H \cong G[U]\,]$ is always an upper bound for p_H, since p_H is the measure of the union of all those events. For example, if $H = \overline{K^k}$, we have the following trivial upper bound on the probability that G contains an induced copy of H:

[11.2.1] [11.3.4] **Lemma 11.1.2.** *For all integers n, k with $n \geqslant k \geqslant 2$, the probability that $G \in \mathcal{G}(n,p)$ has a set of k independent vertices is at most*

$$\mathbb{P}[\,\alpha(G) \geqslant k\,] \leqslant \binom{n}{k} q^{\binom{k}{2}}.$$

Proof. The probability that a fixed k-set $U \subseteq V$ is independent in G is $q^{\binom{k}{2}}$. The assertion thus follows from the fact that there are only $\binom{n}{k}$ such sets U. □

Analogously, the probability that $G \in \mathcal{G}(n,p)$ contains a K^k is at most

$$\mathbb{P}[\,\omega(G) \geqslant k\,] \leqslant \binom{n}{k} p^{\binom{k}{2}}.$$

Now if k is fixed, and n is small enough that these bounds for the probabilities $\mathbb{P}[\,\alpha(G) \geqslant k\,]$ and $\mathbb{P}[\,\omega(G) \geqslant k\,]$ sum to less than 1, then $\mathcal{G}$ contains graphs that have neither property: graphs which contain neither a K^k nor a $\overline{K^k}$ induced. But then any such n is a lower bound for the Ramsey number of k!

As the following theorem shows, this lower bound is quite close to the upper bound of 2^{2k-3} implied by the proof of Theorem 9.1.1:

Theorem 11.1.3. (Erdős 1947)
For every integer $k \geqslant 3$, the Ramsey number of k satisfies

$$R(k) > 2^{k/2}.$$

Proof. For $k = 3$ we trivially have $R(3) \geqslant 3 > 2^{3/2}$, so let $k \geqslant 4$. We show that, for all $n \leqslant 2^{k/2}$ and $G \in \mathcal{G}(n, \frac{1}{2})$, the probabilities $\mathbb{P}[\,\alpha(G) \geqslant k\,]$ and $\mathbb{P}[\,\omega(G) \geqslant k\,]$ are both less than $\frac{1}{2}$.

Since $p = q = \frac{1}{2}$, Lemma 11.1.2 and the analogous assertion for $\omega(G)$ imply the following for all $n \leqslant 2^{k/2}$ (use that $k! > 2^k$ for $k \geqslant 4$):

$$\begin{aligned}\mathbb{P}[\,\alpha(G) \geqslant k\,],\ \mathbb{P}[\,\omega(G) \geqslant k\,] &\leqslant \binom{n}{k}\left(\tfrac{1}{2}\right)^{\binom{k}{2}} \\ &< \left(n^k/2^k\right) 2^{-\frac{1}{2}k(k-1)} \\ &\leqslant \left(2^{k^2/2}/2^k\right) 2^{-\frac{1}{2}k(k-1)} \\ &= 2^{-k/2} \\ &< \tfrac{1}{2}\,.\end{aligned}$$

□

In the context of random graphs, most of the familiar graph invariants (like average degree, connectivity, chromatic number, and so on) may be interpreted as a non-negative *random variable* on $\mathcal{G}(n,p)$, a function

random variable

$$X \colon \mathcal{G}(n,p) \to [0,\infty)\,.$$

The *mean* or *expected* value of X is the number

mean

expectation

$$\mathbb{E}(X) := \sum_{G \in \mathcal{G}(n,p)} \mathbb{P}(\{G\}) \cdot X(G)\,.$$

$\mathbb{E}$

If X takes integers as values, we can compute $\mathbb{E}(X)$ alternatively by summing over these values k:

$$\mathbb{E}(X) = \sum_{k \geqslant 1} \mathbb{P}[\,X \geqslant k\,] = \sum_{k \geqslant 1} k \cdot \mathbb{P}[\,X = k\,]\,.$$

Note also that the operator $\mathbb{E}$, the *expectation*, is linear: we have $\mathbb{E}(X+Y) = \mathbb{E}(X) + \mathbb{E}(Y)$ and $\mathbb{E}(\lambda X) = \lambda \mathbb{E}(X)$ for any two random variables X, Y on $\mathcal{G}(n,p)$ and $\lambda \geqslant 0$.

Since our probability spaces are finite, the expectation can often be computed by a simple application of *double counting*, a standard combinatorial technique we met before in the proof of Corollary 4.2.10. For example, if X is a random variable on $\mathcal{G}(n,p)$ that counts the number of subgraphs of G in some fixed set $\mathcal{H}$ of graphs on V, then $\mathbb{E}(X)$, by definition, counts the number of pairs (G,H) such that $H \in \mathcal{H}$ and $H \subseteq G$, each weighted with the probability $\mathbb{P}(\{G\})$. Algorithmically, we compute $\mathbb{E}(X)$ by going through the graphs $G \in \mathcal{G}(n,p)$ in an 'outer loop' and performing, for each G, an 'inner loop' that runs through the graphs $H \in \mathcal{H}$ and counts '$\mathbb{P}(\{G\})$' whenever $H \subseteq G$. Alternatively, we may count the same set of weighted pairs with H in the outer and G in the inner loop. This amounts to adding up, over all $H \in \mathcal{H}$, the probabilities $\mathbb{P}[\,H \subseteq G\,]$:

$$\mathbb{E}(X) = \sum_{G \in \mathcal{G}(n,p)} \bigl|\{\, H \in \mathcal{H} : H \subseteq G \,\}\bigr| \cdot \mathbb{P}(\{G\}) = \sum_{H \in \mathcal{H}} \mathbb{P}[\,H \subseteq G\,].$$

To illustrate this once in detail, and to introduce the probabilistic terminology commonly used at this point, let us compute the expected number of cycles of some given length $k \geqslant 3$ in a random graph $G \in \mathcal{G}(n,p)$. (We shall also need this for our proof of Erdős's theorem in Section 11.2.) Let $X: \mathcal{G}(n,p) \to \mathbb{N}$ be the random variable that assigns to a random graph G its number of k-cycles, the number of subgraphs isomorphic to C^k.

X

How many potential such cycles are there? In other words, how large is the set $\mathcal{C}_k$ of all k-cycles with vertices in V? Since there are

$\mathcal{C}_k$

$(n)_k$

$$(n)_k := n\,(n-1)(n-2)\cdots(n-k+1)$$

ways of choosing a sequence of k distinct vertices in V, and each k-cycle is identified by $2k$ of those sequences, clearly

$$|\mathcal{C}_k| = (n)_k/2k\,. \qquad (1)$$

[11.2.2] [11.4.3]

Lemma 11.1.4. *The expected number of k-cycles in $G \in \mathcal{G}(n,p)$ is*

$$\mathbb{E}(X) = \frac{(n)_k}{2k}\,p^k.$$

Proof. Consider for every fixed $C \in \mathcal{C}_k$ its *indicator random variable* $X_C: \mathcal{G}(n,p) \to \{0,1\}$, defined by

$$X_C\colon\ G \mapsto \begin{cases} 1 & \text{if } C \subseteq G; \\ 0 & \text{otherwise.} \end{cases}$$

Since X_C takes only 1 as a positive value, its expectation $\mathbb{E}(X_C)$ equals the measure $\mathbb{P}\,[\,X_C = 1\,]$ of the set of all graphs in $\mathcal{G}(n,p)$ that contain C. But this is just the probability that $C \subseteq G$:

$$\mathbb{E}(X_C) = \mathbb{P}\,[\,C \subseteq G\,] = p^k. \qquad (2)$$

Our random variable X assigns to every graph G its number of k-cycles. Hence

$$X(G) = \sum_{C \in \mathcal{C}_k} X_C(G)$$

for every G, or $X = \sum X_C$ for short. Since the expectation is linear, applying this with (1) and (2) yields

$$\mathbb{E}(X) = \sum_{C \in \mathcal{C}_k} \mathbb{E}(X_C) = \sum_{C \in \mathcal{C}_k} \mathbb{P}\,[\,C \subseteq G\,] = \frac{(n)_k}{2k}\,p^k$$

as claimed. □

Computing the mean of a random variable X can be a simple and effective way to establish the existence of a graph G such that $X(G) < a$ for some fixed $a > 0$ and, moreover, G has some desired property $\mathcal{P}$. Indeed, if the expected value of X is small, then $X(G)$ cannot be large for more than a few graphs in $\mathcal{G}(n,p)$, because $X(G) \geqslant 0$ for all $G \in \mathcal{G}(n,p)$. Hence X must be small for many graphs in $\mathcal{G}(n,p)$, and it is reasonable to expect that among these we may find one with the desired property $\mathcal{P}$.

This simple idea lies at the heart of countless non-constructive existence proofs using random graphs, including the proof of Erdős's theorem presented in the next section. Quantified, it takes the form of the following lemma, whose proof follows at once from the definition of the expectation and the additivity of $\mathbb{P}$:

[11.2.2]
[11.4.1]
[11.4.3]

Lemma 11.1.5. (Markov's Inequality)
Let $X \geqslant 0$ be a random variable on $\mathcal{G}(n,p)$ and $a > 0$. Then

$$\mathbb{P}[\,X \geqslant a\,] \leqslant \mathbb{E}(X)/a\,.$$

Proof.

$$\mathbb{E}(X) = \sum_{G \in \mathcal{G}(n,p)} \mathbb{P}(\{G\}) \cdot X(G) \;\geqslant \sum_{\substack{G \in \mathcal{G}(n,p) \\ X(G) \geqslant a}} \mathbb{P}(\{G\}) \cdot a = \mathbb{P}[\,X \geqslant a\,] \cdot a\,.$$

□

11.2 The probabilistic method

Very roughly, the *probabilistic method* in discrete mathematics has developed from the following idea. In order to prove the existence of an object with some desired property, one defines a probability space on some larger – and certainly non-empty – class of objects, and then shows that an element of this space has the desired property with positive probability. The 'objects' inhabiting this probability space may be of any kind: partitions or orderings of the vertices of some fixed graph arise as naturally as mappings, embeddings and, of course, graphs themselves. In this section, we illustrate the probabilistic method by giving a detailed account of one of its earliest results: of Erdős's classic theorem on large girth and chromatic number (Theorem 5.2.5).

Erdős's theorem says that, given any positive integer k, there is a graph G with girth $g(G) > k$ and chromatic number $\chi(G) > k$. Let us call cycles of length at most k *short*, and sets of $|G|/k$ or more vertices *big*. For a proof of Erdős's theorem, it suffices to find a graph G without short cycles and without big independent sets of vertices: then the colour classes in any vertex colouring of G are *small* (not big), so we need more than k colours to colour G.

short
big/small